POWER PULSE

Power Cables Important Formulae

Notes by Balaji Sir

1 Single-core cable (conductor radius r, sheath inner radius R)

1.

$$C = \frac{2\pi\varepsilon_0\varepsilon_r}{\ln(R/r)}\ \mathrm{F/m} = \frac{0.0241\,\varepsilon_r}{\log_{10}(R/r)}\ \mu\mathrm{F/km}$$

capacitance per unit length

2.

$$I_c = 2\pi f\, C\, V_{ph} \qquad \text{Charging kVA} = 3\,V_{ph} I_c$$

charging current of a 3-phase cable

3.

$$R_{ins} = \frac{\rho}{2\pi\ell}\ln\frac{R}{r}$$

insulation resistance of length ℓ; ρ = resistivity of insulation; it falls as length increases

4.

$$g(x) = \frac{V}{x\ln(R/r)}, \qquad g_{max} = \frac{V}{r\ln(R/r)}, \quad g_{min} = \frac{V}{R\ln(R/r)}$$

dielectric stress at radius x; V = peak voltage between conductor and sheath; g_max at the conductor surface

5.

$$r = \frac{R}{e} \ \Rightarrow\ g_{max} = \frac{eV}{R}\ \text{(minimum)}$$

most economical conductor radius for a given R and V

2 Grading

6.

$$\varepsilon_1 r_1 = \varepsilon_2 r_2 = \varepsilon_3 r_3 = \cdots$$

capacitance grading: equal maximum stress in every layer

7.

$$V = g_{max}\left[r_1\ln\frac{r_2}{r_1} + r_2\ln\frac{r_3}{r_2} + r_3\ln\frac{R}{r_3}\right]$$

voltage with three layers and equal maximum stress

3 Three-core belted cable

8.

$$C_N = C_s + 3C_c = 2\,C_{ab}\ \text{(per phase to neutral)}$$

C_c = capacitance between cores, C_s = core to sheath; C_ab = measured between two cores with the third connected to the sheath

9.

$$C_{\text{bunched cores–sheath}} = 3C_s$$

4 Losses and heating

10.

$$W_d = 2\pi f\, C\, V^2 \tan\delta\ \text{(per phase)}$$

dielectric loss; V = phase voltage

11.

$$R_{th} = \frac{\rho_{th}}{2\pi}\ln\frac{R}{r}\ (^\circ\mathrm{C\cdot m/W}), \qquad \Delta T = Q\,R_{th}$$

thermal resistance of the insulation per unit length; Q = heat flow per unit length